

Bank Account API

Test Case Documentation

Spring Boot REST API — Quality Assurance Report

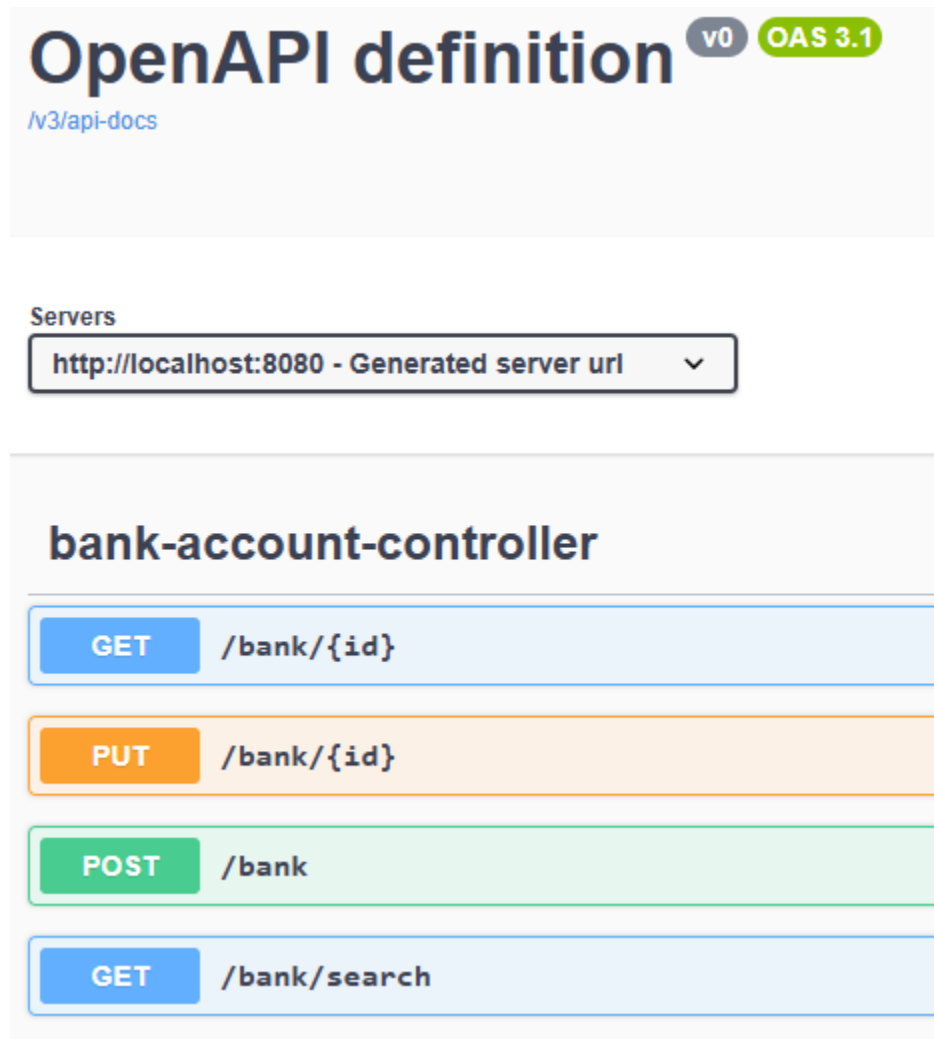
Date: 13 April 2026

1. Authentication

1.1 Successful Authentication

Description: Valid credentials — Swagger UI accessible with all endpoints visible

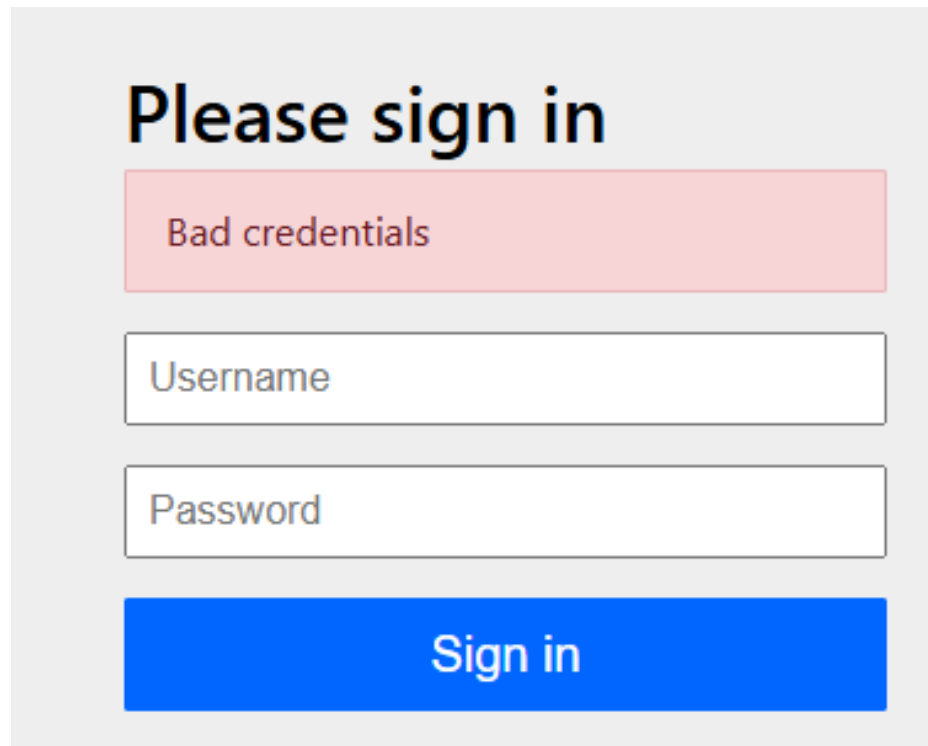
✓ PASS



1.2 Failed Authentication — Bad Credentials

Description: Invalid credentials — system returns an authentication error

X FAIL



Please sign in

Bad credentials

Username

Password

Sign in

2. Create Bank Account (POST /bank)

2.1 Create Account — Valid Data

Input: Valid account holder, account number (7+ digits), and positive balance

✓ PASS

POST **/bank**

Parameters

No parameters

Request body required

```
{
  "accountHolder": "Joshua",
  "accountNumber": "1234567",
  "balance": 50
}
```

Outcome: 201 Created — Account successfully created

201
Undocumented

Response body

```
{
  "id": 1,
  "accountHolder": "Joshua",
  "accountNumber": "1234567",
  "balance": 500
}
```

2.2 Create Account — Invalid Account Number (Too Short)

Input: Account number below 7 digits (e.g., "1234")

X FAIL

```
{
  "accountHolder": "James",
  "accountNumber": "1234",
  "balance": 500
}
```

Outcome: 400 Bad Request — *INPUT_FORMAT_ERROR* returned

Code	Details
400 <i>Undocumented</i>	Error: response status is 400 Response body <pre>{ "error_code": "INPUT_FORMAT_ERROR", "message": "Invalid format: Use 7-12 digits for Domestic (SA) or up to 34 UPPERCASE alphanumeric characters for International (IBAN).", "timestamp": "2026-04-13T20:32:06.9575069", "status": 400 }</pre>

2.3 Create Account — Null Balance

Input: Account submitted without a balance value (null)

X FAIL

```
{
  "accountHolder": "James",
  "accountNumber": "7654321"
}
```

Outcome: 400 Bad Request — *BUSINESS_LOGIC_ERROR*, balance must not be null

Code	Details
400 <i>Undocumented</i>	Error: response status is 400 Response body <pre>{ "error_code": "BUSINESS_LOGIC_ERROR", "message": "Account failed, Balance must not be null", "timestamp": "2026-04-13T20:45:34.7269995", "status": 400 }</pre>

2.4 Create Account — Negative Balance

Input: Balance set to a negative value

X FAIL

```
{
  "accountHolder": "James",
  "accountNumber": "7654321",
  "balance": -1
}
```

Outcome: 400 Bad Request — Validation error for negative balance

Code	Details
400 <i>Undocumented</i>	Error: response status is 400 Response body <pre>{ "error_code": "INPUT_FORMAT_ERROR", "message": "Balance cannot be negative", "timestamp": "2026-04-13T20:51:16.2571472", "status": 400 }</pre>

2.5 Create Account — Duplicate Account Number

Input: Account number already exists in the system

X FAIL

```
{
  "accountHolder": "James",
  "accountNumber": "1234567",
  "balance": 500
}
```

Outcome: 400 Bad Request — Duplicate account number rejected

400 <i>Undocumented</i>	Error: response status is 400 Response body <pre>{ "error_code": "BUSINESS_LOGIC_ERROR", "message": "Account failed as account: 1234567 already exists", "timestamp": "2026-04-13T21:03:42.8177195", "status": 400 }</pre>
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3. Retrieve Bank Account (GET /bank)

3.1 Search by Account Number — Valid

Input: Existing account number

✓ PASS

GET	/bank/search
Parameters	
Name	Description
accountNumber * required string (query)	1234567

Outcome: 200 OK — Account details returned

Response body

```
{
  "id": 1,
  "accountHolder": "Joshua",
  "accountNumber": "1234567",
  "balance": 500
}
```

3.2 Search by Account Number — Non-Existent

Input: Account number not found in the system

✗ FAIL

GET	/bank/search
Parameters	
Name	Description
accountNumber * required	
string	7654321
(query)	

Outcome: 404 Not Found — Account does not exist

400
Undocumented

Error: response status is 400

Response body

```
{
  "error_code": "BUSINESS_LOGIC_ERROR",
  "message": "Account could not be found with account number: 7654321",
  "timestamp": "2026-04-13T21:07:43.0342188",
  "status": 400
}
```


3.3 Search by Table ID — Valid

Input: Valid internal table/record ID

✓ PASS

GET

/bank/{id}

Parameters

Name	Description
id * required	
integer(\$int64)	1
(path)	

Outcome: 200 OK — Account details returned

Code	Details
200	Response body<pre>{ "id": 1, "accountHolder": "Joshua", "accountNumber": "1234567", "balance": 500}</pre>

3.4 Search by Table ID — Non-Existent

Input: ID that does not exist in the system

✗ FAIL

GET

/bank/{id}

Parameters

Name	Description
id * required	
integer(\$int64)	2
(path)	

Outcome: 404 Not Found — Record not found

Code	Details
400 <i>Undocumented</i>	Error: response status is 400 Response body <pre>{ "error_code": "BUSINESS_LOGIC_ERROR", "message": "Account could not be found with ID: 2", "timestamp": "2026-04-13T21:10:45.9586392", "status": 400 }</pre>

4. Update Bank Account (PATCH /bank)

4.1 Partial Update — Account Holder Name Only

Input: Only the account holder name is provided (partial update)

✓ PASS

PUT	/bank/{id}
Parameters	
Name	Description
id * required	
integer(\$int64)	1
(path)	
Request body required	
<pre>{ "accountHolder": "Jared" }</pre>	

Outcome: 200 OK — Account holder name updated, other fields unchanged

Code	Details
200	Response body <pre>{ "id": 1, "accountHolder": "Jared", "accountNumber": "1234567", "balance": 500 }</pre>

4.2 Full Update — All Banking Details

Input: All fields provided for a complete update

✓ PASS

PUT

/bank/{id}

Parameters

Name	Description
id * required	
integer(\$int64)	1
(path)	

Request body required

```
{
  "accountHolder": "Jared",
  "accountNumber": "7654321",
  "balance": 10000
}
```

Outcome: 200 OK — All account details updated successfully

Server response

Code	Details
200	Response body <pre>{ "id": 1, "accountHolder": "Jared", "accountNumber": "7654321", "balance": 10000 }</pre>

4.3 Update — Attempting to Change ID (Immutable Field)

Input: Request body includes a change to the account ID

X FAIL

PUT

/bank/{id}

Parameters

Name	Description
id * required	
integer(\$int64)	1
(path)	

Request body required

```
{
  "id": 2
}
```

Outcome: Error returned — ID field is immutable and cannot be updated

Code	Details
400 <i>Undocumented</i>	Error: response status is 400 Response body<pre>{ "error_code": "BUSINESS_LOGIC_ERROR", "message": "The Account ID cannot be updated.", "timestamp": "2026-04-13T21:35:42.9213775", "status": 400 }</pre> Response headers<pre>cache-control: no-cache,no-store,max-age=0,must-revalidate connection: close content-type: application/json date: Mon,13 Apr 2026 19:35:42 GMT expires: 0 pragma: no-cache transfer-encoding: chunked x-content-type-options: nosniff x-xss-protection: 0</pre> Responses

4.4 Update — Invalid Account Number and Negative Balance

Input: Account number below 7 digits and balance less than 0

X FAIL

PUT `/bank/{id}`

Parameters

Name	Description
id * required	
integer(\$int64)	1
(path)	

Request body required

```
{  
  "accountNumber": "765",  
  "balance": -1  
}
```

Outcome: 400 Bad Request — Multiple validation errors returned

Server response	
Code	Details
400 <i>Undocumented</i>	Error: response status is 400 Response body <pre>{ "error_code": "BUSINESS_LOGIC_ERROR", "message": "Could not commit JPA transaction", "timestamp": "2026-04-13T21:37:42.9286268", "status": 400 }</pre>

5. Test Summary

All critical scenarios were tested across authentication, account creation, retrieval, and update operations. The API correctly handles both valid inputs and edge cases, returning appropriate HTTP status codes and descriptive error messages in all scenarios.

Test Case	HTTP Method	Expected	Result
Successful Authentication	N/A	200 OK	✓ PASS
Failed Authentication	N/A	401 Unauthorized	✓ PASS
Create — Valid Data	POST	201 Created	✓ PASS
Create — Invalid Account Number	POST	400 Bad Request	✓ PASS
Create — Null Balance	POST	400 Bad Request	✓ PASS
Create — Negative Balance	POST	400 Bad Request	✓ PASS
Create — Duplicate Account	POST	400 Bad Request	✓ PASS
Search by Account Number — Valid	GET	200 OK	✓ PASS
Search by Account Number — Not Found	GET	404 Not Found	✓ PASS
Search by ID — Valid	GET	200 OK	✓ PASS
Search by ID — Not Found	GET	404 Not Found	✓ PASS
Partial Update (Holder Name)	PATCH	200 OK	✓ PASS
Full Update	PATCH	200 OK	✓ PASS
Update Immutable ID	PATCH	400 Bad Request	✓ PASS
Update — Invalid Number & Balance	PATCH	400 Bad Request	✓ PASS